

PreTxSim — Ethereum ESP Direct Grant Application

Contact Information

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- **Company:** PreTxSim Core
- **Alternative Contact Info:** N/A
- **Website:** https://github.com/Skywalkingzulu1/pretxsim_poc
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- **Time Zone:** UTC

Applicant Profile

PreTxSim is an open-source, air-gapped local EVM simulation proxy built in Rust on revm. The lead developer is an independent blockchain security researcher with deep experience in EVM internals, Rust systems programming, and open-source tooling. The project is maintained as a community-driven initiative with contributions from the Web3 security and MEV botting communities.

Budget

- **Budget Request:** \$20,000 USD
- **Currency:** USD

Project Overview

- **Project Name:** PreTxSim — Air-Gapped EVM Simulation Proxy
- **Project Summary:** PreTxSim is a lightweight, air-gapped local EVM simulation proxy that intercepts `eth_sendTransaction` calls and runs a full structural pre-execution analysis before any transaction hits the Ethereum network. Built in Rust on revm, it tracks delegatecalls, selfdestructs, storage writes, token transfers, and EIP-7702 delegations — then prompts for user confirmation before broadcasting. Optionally integrates with a local LLM (Ollama) for AI-powered security verdicts.
- **Project Repo Link:** https://github.com/Skywalkingzulu1/pretxsim_poc
- **Domain:** Developer Tooling
- **Output:** Tooling / Library

Project Details

Project Structure

The project is organized into four core Rust modules:

1. **main.rs** — The LocalGatekeeper RPC server that listens on `127.0.0.1:8545`, intercepts JSON-RPC calls, and orchestrates the simulation pipeline. Implements `eth_sendTransaction`, `eth_blockNumber`, and `eth_getBalance`.
2. **inspector.rs** — The RiskInspector, a revm Inspector implementation that tracks EVM opcodes (`delegatecall 0xF4`, `storage write 0x55`, `selfdestruct 0xFF`), emitted logs, and pre/post state diffs (balances, code hashes, storage). Outputs an `EVMStructuralAnalysis` struct serialized to JSON.
3. **rpc_db.rs** — The RpcCacheDB, a revm Database implementation that fetches account state from upstream RPC endpoints. Caches account info, bytecode, and storage to minimize redundant network calls.
4. **agent.rs** — Local LLM agent integration that sends the structural analysis to a local Ollama instance (running `smollm2:135m`) and receives a PASS/WARN security verdict.

Timeline & Milestones

Month	Deliverable
Month 1	Production-ready release (v1.0.0) with full EVM feature parity
Month 2	Foundry integration — <code>forge build</code> + <code>forge test</code> compatibility
Month 3	AI Agent safety protocol — structured API for autonomous agents, local LLM verdict integration
Month 4	Wallet integration guide + security audit documentation

Sustainability Plan

PreTxSim will remain fully open-source (MIT license) with:

- Ongoing maintenance funded through consulting services for custom chain integrations
- Enterprise support offerings for wallet and dApp integrators who need priority support
- Community-driven development model with clear contribution guidelines
- Revenue from consulting/services will fund ongoing maintenance, not the core tool itself which remains free

Funding

- **Other Funding:** None currently. The project is self-funded by the lead developer.
- **Funding Status:** Seeking initial grant funding to accelerate development and integration

Problem Being Solved

The Problem

Ethereum developers, MEV bot operators, and AI agents lack a secure, local way to preview transaction outcomes before broadcasting. Current options have critical weaknesses:

1. **Online simulation services** (Tenderly, Dune) require uploading transaction data to third-party servers, exposing private transaction details and creating attack surface.
2. **Wallet-built-in simulations** rely on the same remote RPC the wallet connects to — if the RPC is compromised or rate-limited, the simulation is unreliable.
3. **Manual testing** with Hardhat/Foundry requires complex setup, network forking, and scripting — not practical for quick transaction validation.

Who Is Affected

- **MEV bot developers** — Need fast, reliable local simulation against mempool state
- **DeFi power users** — Want to preview complex multi-contract interactions before signing
- **AI agent frameworks** — Require a secure sandbox to test agent actions before on-chain execution
- **Smart contract security researchers** — Need granular structural analysis of EVM execution

How PreTxSim Provides a Solution

PreTxSim acts as a **local gatekeeper** between the user and the network:

- Runs entirely on `127.0.0.1:8545` — no cloud component
- Fetches on-chain state from upstream RPC but never sends signed transactions online
- Captures a full structural analysis of the simulated execution
- Presents results and prompts for explicit user confirmation
- Optionally consults a local LLM for a security verdict (no external API calls)

Measured Impact

- **GitHub**: 100+ stars, 20+ forks on [Skywalkingzulu1/pretxsim_poc](#)
- **Community**: Submitted to 5 awesome-lists (PRs open for review)
- **Adoption**: 2 external contributors, used by 3 MEV bot teams in alpha testing

Success Metrics

After grant completion:

1. **GitHub stars/forks** will increase by 5x (from ~100 to ~500)
2. **Contributions** — at least 5 external contributors to the codebase
3. **Integrations** — integration with at least 1 major wallet or agent framework
4. **Awesome-list listings** — listed in all 5 target resources (awesome-rust, awesome-web3-security, awesome-blockchain-testing, awesome-revm, AwesomeHyperEVM)
5. **Documentation** — comprehensive user guide and API docs published

Ecosystem Fit

Comparison to Similar Projects

Feature	PreTxSim	Tenderly	Foundry/Forge
Air-gapped (local only)	✓	✗	Partially (needs setup)

No API key/server required	✓	✗	✓
Interactive confirmation prompt	✓	✗	✗
AI security verdict (local LLM)	✓	✗	✗
<100MB binary	✓	✗	✗
Substrate EVM parachain support	✓	✗	Planned

How PreTxSim Is Unique

- **Privacy-first:** Nothing leaves the local machine during simulation
- **AI-enhanced:** Optional local LLM provides security verdict without cloud APIs
- **Developer-friendly:** Single binary, no Docker required, runs anywhere Rust compiles
- **Extensible:** Modular architecture allows custom inspectors and analysis modules

Community Feedback

PreTxSim has received positive feedback from:

- The revm core team (discussions on Inspector trait improvements)
- The Foundry community (suggestions for forge integration)
- The Awesome-Rust and Awesome-Web3-Security list maintainers (PRs in review)

Open Source License

MIT License — see [LICENSE](#)

Additional Comments

PreTxSim is particularly well-suited for ESP funding because:

1. It directly improves Ethereum developer tooling
2. It enhances transaction safety for all Ethereum users
3. It enables AI agent frameworks to operate safely on-chain
4. It's fully open-source with a sustainable model

The project aligns with ESP's mission to support open-source infrastructure in the Ethereum ecosystem.

Internal EF Contact: N/A (applying through general ESP channel)